

14.2 Phase Changes

Question Paper

Course	CIE A Level Physics
Section	14. Temperature
Topic	14.2 Phase Changes
Difficulty	Medium

Time allowed: 60

Score: /46

Percentage: /100

Question 1a

Compare *freezing* and *condensation*.

[2 marks]

Question 1b

The effective power of an oven is 1500 W.

The temperature of a mass of 160 g of water inside some food rises from 15 °C to 90 °C.

Calculate the time taken to heat the food to this temperature.

Specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$

time = minutes

[3 marks]

Question 1c

A student conducts an experiment into the heating of water inside food in an oven.

They calculate the specific heat capacity of water and obtain a higher value than that given in part **(b)**.

Suggest a reason for this difference.

[1 mark]

Question 2a

Explain the statement *the specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$* .

[2 marks]

Question 2b

The heater in an electric kettle has a power of 3.18 kW.

When the water in the kettle is boiling at a steady rate, the mass of evaporated water in 1.5 minutes and the rate of loss of thermal energy to the surroundings is 500 W. The specific latent heat of vaporisation of water is 2.26 MJ kg^{-1} .

Show that 107 g of water is evaporated in this time.

[4 marks]**Question 2c**

A student places water in an ice cube tray of a freezer to calculate the specific latent heat of fusion of ice.

700 g of water is placed in the ice cube tray at 0°C and the freezer cools them to -3°C . A total of 0.23 MJ of energy is removed from the water to turn it into ice at -3°C .

Specific heat capacity of ice = $2.1 \text{ kJ kg}^{-1} \text{ K}^{-1}$.

Calculate the specific latent heat of fusion of ice.

[3 marks]

Question 2d

Suggest why the specific latent heat of fusion of water is much less than the specific latent heat of vaporisation of water.

[2 marks]

Question 3a

The energy required to boil water at $100\text{ }^{\circ}\text{C}$ to steam at $100\text{ }^{\circ}\text{C}$ is much greater than the energy required to melt ice at $0\text{ }^{\circ}\text{C}$ to water at $0\text{ }^{\circ}\text{C}$.

Explain this, in terms of the spacing of the molecules.

[1 mark]

Question 3b

A heater supplies energy at a constant rate to 64 g of a substance. The variation with time of the temperature of the substance is shown in Fig. 1.1. The substance is perfectly insulated from its surroundings.

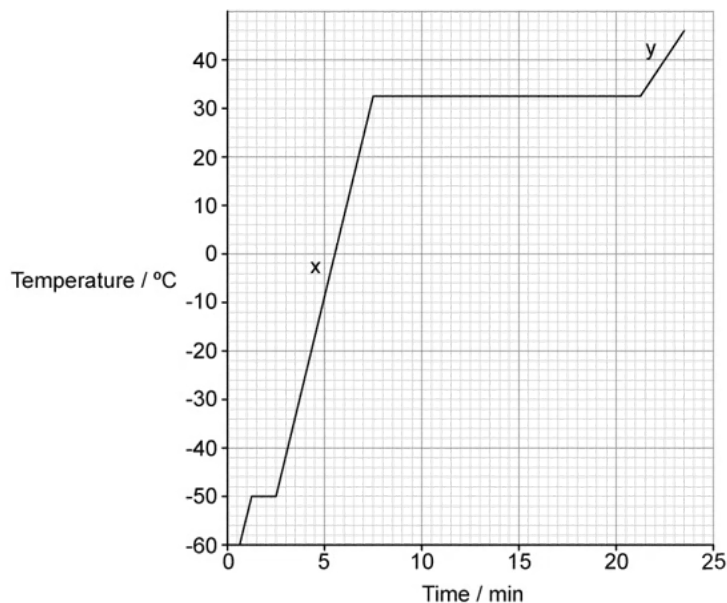


Fig 1.1

The power of the heater is 230 W.

Use data from Fig. 1.1 to calculate, in MJ kg^{-1} , the specific latent heat of vaporisation, L of the substance.

[3 marks]

Question 3c

Calculate the ratio $\frac{\text{specific latent heat of vaporisation}}{\text{specific latent heat of fusion}}$.

[3 marks]

Question 3d

Suggest what can be deduced from the fact that section **X** on the graph is steeper than section **Y** in Fig 1.1.

[1 mark]

Question 4a

Calculate the energy released when 2.5 kg of water at 17 °C cools to 0 °C and then freezes to form ice, also at 0 °C.

Specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific latent heat of fusion of ice = $3.4 \times 10^5 \text{ J kg}^{-1}$

[4 marks]

Question 4b

Explain why it is more effective to cool ice packs by placing them in a box full of melting ice rather than in a box of water at an initial temperature of 0 °C.

[2 marks]

Question 4c

The silica gel and water solution from the inside of an ice pack of mass 0.450 kg at a temperature of 2.00 °C are emptied and poured into a plastic beaker. The beaker has a mass of 0.210 kg and is initially at a temperature of 19.0 °C.

Specific heat capacity of plastic = 1670 J kg⁻¹ K⁻¹

Specific heat capacity of silica gel and water = 6420 J kg⁻¹ K⁻¹

Calculate the final temperature, T_f , of the silica gel and water solution when it reaches thermal equilibrium with the beaker.

Assume no heat is gained from or lost to the surroundings.

[2 marks]

Question 4d

The silica gel and water solution, along with the beaker are cooled from T_f to a temperature of 5 °C by adding ice at a temperature of 0 °C.

Calculate the mass of ice added.

Assume no heat is gained from or lost to the surroundings.

Specific latent heat of fusion of ice = 3.34×10^5 J kg⁻¹

[3 marks]

Question 5a

Explain the meaning of the statement *the specific heat capacity of water is 4200 J kg⁻¹ K⁻¹*.

[2 marks]

Question 5b

An engineer is designing an ice-making machine. Water will enter the device at 15°C and the ice cubes are to be cooled to -6°C before release.

Show that about 0.3 MJ of energy must be removed from 0.75 kg of water at 15°C to change it into ice at -6°C .

Specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific heat capacity of ice = $2.1 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific latent heat of fusion of ice = $3.3 \times 10^5 \text{ J kg}^{-1}$

[4 marks]**Question 5c**

The design brief requires that 1.25 kg of water is frozen in 240 s.

Calculate the rate at which energy must be removed from the machine.

[2 marks]**Question 5d**

The specific latent heat of vaporisation of water is $2.3 \times 10^6 \text{ J kg}^{-1}$.

Suggest why the specific latent heat of vaporisation of water is much greater than the specific latent heat of fusion of water.

[2 marks]

Model Answers: Medium

1a

(a) Comparing freezing and condensation:

- Freezing is from liquid to solid **AND** at a specific / one temperature / at the melting point; [1 mark]
- Condensation is vapour to liquid **AND** at all temperatures; [1 mark]

[Total: 2 marks]

1b

(b) The time taken to heat the food to this temperature is:

List the known quantities:

- Power of the oven, $P = 1500 \text{ W}$
- Mass of water, $m = 160 \text{ g} = 160 \times 10^{-3} \text{ kg}$
- Initial temperature, $\theta_i = 15^\circ \text{C}$
- Final temperature, $\theta_f = 90^\circ \text{C}$
- Specific heat capacity of water, $c = 4200 \text{ J kg}^{-1} \text{K}^{-1}$

Determine the equation to calculate the time taken to heat the food:

- $\Delta Q = Pt = mc\Delta\theta$ [1 mark]
- $t = \frac{mc\Delta\theta}{P}$

Substitute in the values:

- $t = \frac{(160 \times 10^{-3}) \times 4200 \times (90 - 15)}{1500}$ [1 mark]
- $t = 33.6 \text{ s}$

Convert time into minutes:

- $t = \frac{33.6}{60} = 0.56 \text{ minutes}$ [1 mark]

[Total: 3 marks]

Remember ΔQ is the **change in heat energy** which can also be related to **power** and **time**. The combination of these equations appear many times in exam questions.

1c

(c) A reason for the difference in the specific heat capacity of water is:

Any **one** from:

- Energy is lost in its heat store from the oven; [1 mark]
- The container containing the food / water is also heated; [1 mark]
- Energy in its heat store heats up the oven door / inside the oven too; [1 mark]

Any sensible suggestion

[Total: 3 marks]

2a

(a) The statement *the specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$* means:

- The energy required to raise 1 kg of water by 1 K (or 1 °C); [1 mark]
- Is 4200 J; [1 mark]

[Total: 2 marks]

2b

(b) Show that 107 g of water is evaporated in this time:

List the known quantities:

- Power of the kettle, $P = 3.18 \text{ kW} = 3180 \text{ W}$
- Time, $t = 1.5 \text{ minutes} = 1.5 \times 60 = 90 \text{ s}$
- Rate of thermal energy lost to surroundings = 500 W
- The specific latent heat of vaporisation of water is $2.26 \text{ MJ kg}^{-1} = 2.26 \times 10^6 \text{ J kg}^{-1}$

Calculate the energy supplied by the kettle:

- $E = Pt = 3180 \times 90 = 286\,200 \text{ J}$ [1 mark]

Determine the equation for the energy required to evaporate the water, Q :

- Specific latent heat of vapourisation, $Q = mL$

Determine the equation for the mass of the water evaporated, m :

- *thermal energy lost to surroundings* $= E - Q$ [1 mark]

- *rate of thermal energy lost* $\times t = E - mL$

- $m = \frac{E - (\text{rate of thermal energy lost} \times t)}{L}$

Substitute in the values:

- $m = \frac{286\,200 - (500 \times 90)}{2.26 \times 10^6}$ [1 mark]

- $m = 0.1067 \text{ kg} = 107 \text{ g}$ [1 mark]

[Total: 4 marks]

The wording of this question is what makes it difficult. Remember that power is just 'the **rate of transfer of energy**'. The equation energy $E = Pt$ is used a lot in the specific heat capacity exam questions.

The energy transferred from the power provided by the kettle goes to 1. boiling the water (using latent heat of vaporisation) and 2. thermal energy to the surroundings.

2c

(c) Calculate the specific latent heat of fusion of ice:

List the known quantity:

- Mass of water, $m = 700 \text{ g} = 700 \times 10^{-3} \text{ kg}$
- Initial temperature of the ice, $\theta_i = 0^\circ \text{C}$
- Final temperature of the ice, $\theta_f = -4^\circ \text{C}$
- Total energy removed from the water, $E_T = 0.23 \text{ MJ} = 0.23 \times 10^6 \text{ J}$
- Specific heat capacity of ice, $c = 2.1 \text{ kJ kg}^{-1} \text{ K}^{-1} = 2.1 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Determine the equation of the total energy removed from the water:

- Total energy removed = energy to freeze water into ice + energy to cool ice from 0°C to -4°C
- Total energy removed = $mL + mc\Delta\theta$ [1 mark]

Rearrange the equation for the specific latent heat of ice, L :

- $$L = \frac{\text{total energy removed} - mc\Delta\theta}{m}$$

Substitute in the values:

- $$L = \frac{(0.23 \times 10^6) - [(700 \times 10^{-3}) \times (2.1 \times 10^3) \times (-4 - 0)]}{700 \times 10^{-3}}$$
 [1 mark]
- $$L = 3.4 \times 10^5 \text{ J kg}^{-1}$$
 [1 mark]

[Total: 3 marks]

Make sure to split up the situation into a **change of phase** and a **change of temperature**. These require slightly different equations.

2d

(d) The specific latent heat of fusion of water is much less than the specific latent heat of vaporisation of water because:

Any **two** from:

- More energy has to be supplied to separate molecules (liquid to gas) than to break the solid bond between them (solid to liquid); [1 mark]
- For vaporisation, work is done against atmospheric pressure; [1 mark]
- There is a greater change in potential energy from liquid (water) to gas (water vapour) than from solid (ice) to liquid (water); [1 mark]

[Total: 2 marks]

Although you calculated the latent heat of fusion of ice in part (b), this is the **same** as the latent heat of fusion of water. Although it's named 'fusion' this is the energy required to change 1 kg of liquid into solid **or** solid into liquid as well.

3a

- For vaporisation, work is done against atmospheric pressure; [1 mark]
- There is a greater change in potential energy from liquid (water) to gas (water vapour) than from solid (ice) to liquid (water); [1 mark]

[Total: 2 marks]

Although you calculated the latent heat of fusion of ice in part (b), this is the **same** as the latent heat of fusion of water. Although it's named 'fusion' this is the energy required to change 1 kg of liquid into solid **or** solid into liquid as well.

3a

(a) The energy required to boil water at 100 °C to steam at 100 °C is must greater than the energy required to melt ice at 0 °C to water at 0 °C because:

- There is a much greater increase in spacing of the molecules (for evaporation / vaporisation compared with melting / fusion); [1 mark]

[Total: 1 mark]

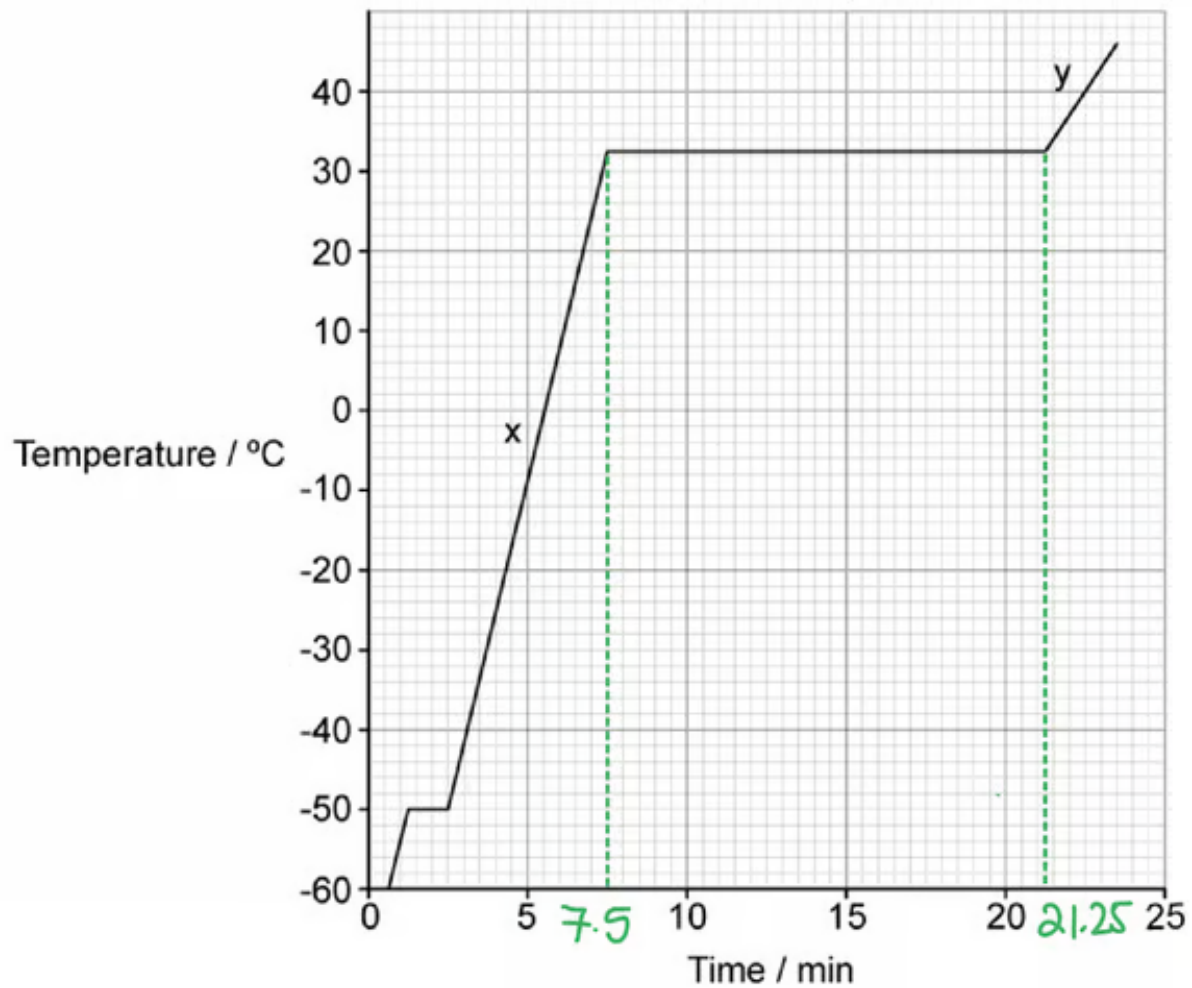
3b

(b) Using data from Fig. 1.1 to calculate, in kJ kg^{-1} , the specific latent heat of vaporisation, L of the substance is:

List the known quantities:

- Power of the heater, $P = 230 \text{ W}$
- Mass of the substance, $m = 64 \text{ g} = 64 \times 10^{-3} \text{ kg}$

Calculate the change in time during vaporisation, Δt :



- $\Delta t = 21.25 - 7.5 = 13.75$ minutes
- $13.75 \times 60 = 825$ s [1 mark]

Determine equation for the specific latent heat of vaporisation, L :

- $\Delta Q = Pt = mL$
- $L = \frac{Pt}{m}$

Substitute in the values:

- $L = \frac{230 \times 825}{64 \times 10^{-3}}$ [1 mark]
- $L = 2.96 \times 10^6 = 3.0 \text{ MJ kg}^{-1}$ [1 mark]

[Total: 3 marks]

If the question asks you to state your final value in a specific unit, make sure to always

convert it into this. You may not get the final mark without it, even if the number is technically correct!

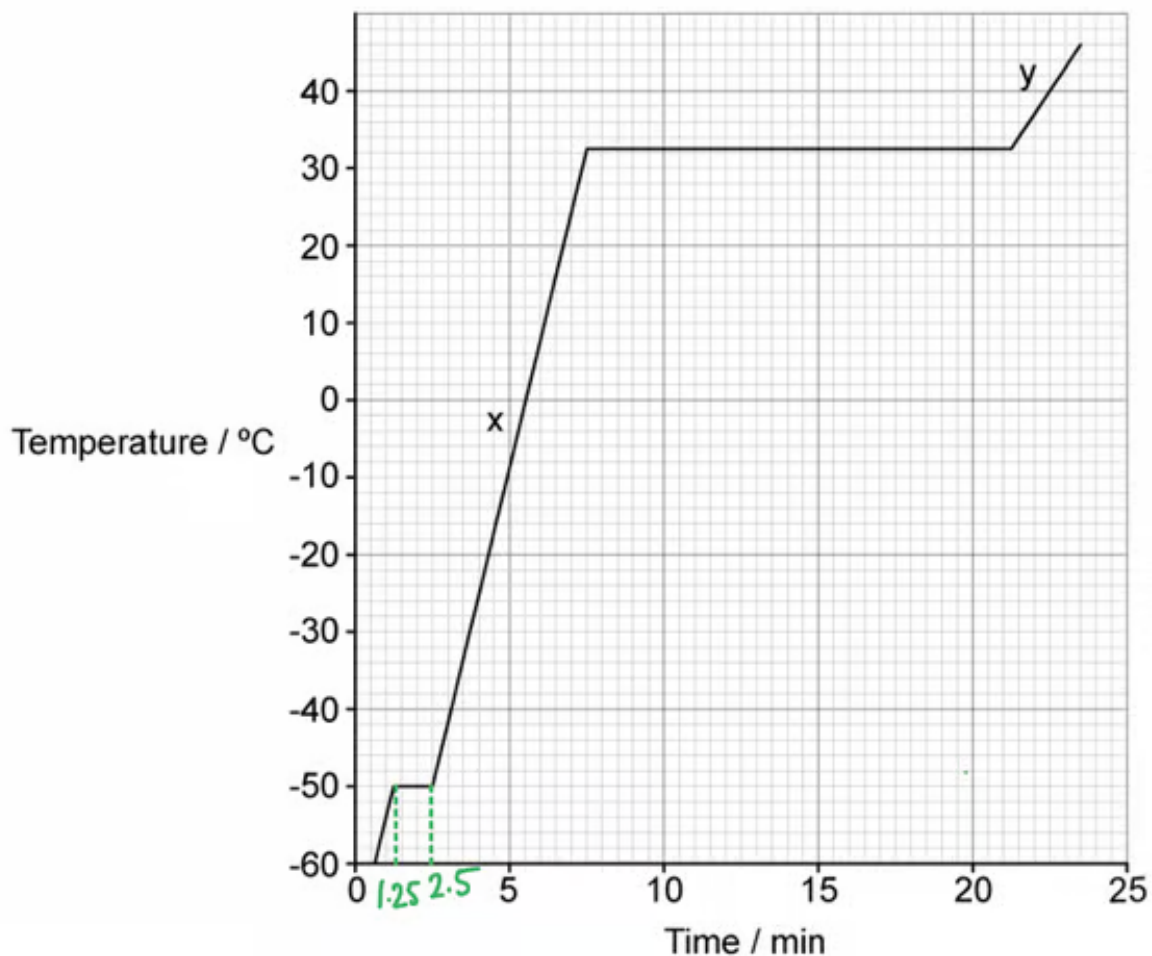
3c

(c) The ratio $\frac{\text{specific latent heat of vaporisation}}{\text{specific latent heat of fusion}}$ is:

List the known quantities:

- Power of the heater, $P = 230 \text{ W}$
- Mass of the substance, $m = 64 \text{ g} = 64 \times 10^{-3} \text{ kg}$
- Specific latent heat of vaporisation, $L_v = 2.96 \times 10^6 \text{ J kg}^{-1}$ (from part b)
- $L = \frac{Pt}{m}$ (from part b)

Calculate the change in time during vaporisation, Δt :



- $\Delta t = 2.5 - 1.25 = 1.25$ minutes
- $1.25 \times 60 = 75$ s [1 mark]

Substitute in the values:

- $L_f = \frac{Pt}{m} = \frac{230 \times 75}{64 \times 10^{-3}}$
- $L_f = 2.96 \times 10^5 = 3.0 \times 10^5 \text{ J kg}^{-1}$ [1 mark]

Calculate the ratio:

- $\frac{L_v}{L_f} = \frac{2.96 \times 10^6}{2.96 \times 10^5} = 10$ [1 mark]

[Total: 3 marks]

Make sure you always convert time into **seconds** when using $E = Pt$ if you want energy E in Joules. This is because power in units of $W = \text{J s}^{-1}$.

3d

(d) The deduction from the fact that section **X** on the graph is steeper than section **Y** is:

- Liquid has a lower specific heat capacity (than gas) **OR** gas has a higher specific heat capacity (than liquid); [1 mark]

[Total: 1 mark]

The gradient of the graph in Fig 1.1 is the **change in temperature with time**. At point X, the substance is in a **liquid** form and it takes less time to increase it by a certain temperature than the graph at point Y does (when it's in gas form).

This means **less** energy is required to change the temperature of the liquid compared to gas. Since specific heat capacity is the energy required to change 1 kg of a substance by 1 °C, this means it has a lower specific heat capacity.

4a

List the known quantities:

3a

(d) The deduction from the fact that section **X** on the graph is steeper than section **Y** is:

- Liquid has a lower specific heat capacity (than gas) **OR** gas has a higher specific heat capacity (than liquid); [1 mark]

[Total: 1 mark]

The gradient of the graph in Fig 1.1 is the **change in temperature with time**. At point X, the substance is in a **liquid** form and it takes less time to increase it by a certain temperature than the graph at point Y does (when it's in gas form).

This means **less** energy is required to change the temperature of the liquid compared to gas. Since specific heat capacity is the energy required to change 1 kg of a substance by 1 °C, this means it has a lower specific heat capacity.

4a

List the known quantities:

- Mass of water, $m_w = 2.5 \text{ kg}$
- Initial temperature of water, $T_i = 17 \text{ °C}$
- Final temperature of water, $T_f = 0 \text{ °C}$
- Specific heat capacity of water, $c_w = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific latent heat of fusion of ice, $c_i = 3.4 \times 10^5 \text{ J kg}^{-1}$

Energy released when the water at 17 °C cools to 0 °C:

- $\Delta Q_{\text{water}} = m_w c_w \Delta \theta_w$
- $\Delta Q_{\text{water}} = 2.5 \times 4200 \times (17 - 0)$ [1 mark]
- $\Delta Q_{\text{water}} = 178\,500 \text{ J}$ [1 mark]

Energy released from changing state from liquid (water) to solid (ice):

- $\Delta Q_{\text{change}} = m_w l_i$
- $\Delta Q_{\text{change}} = 2.5 \times (3.4 \times 10^5) = 850\,000 \text{ J}$ [1 mark]

Total energy released:

- $\Delta Q_{water} + \Delta Q_{change} = 178\,500 + 850\,000 = 1\,028\,500 = 1.03 \times 10^6 \text{ J}$ [1 mark]

[Total: 4 marks]

In specific heat capacity questions, it helps to split the scenario into steps.

In this question, the two steps are: water cools to 0°C → freezes to form ice.

The thermal energy of a substance changes when it is heated or cooled but also when it changes state.

4b

It is more effective to cool ice packs in a box full of ice because:

- Ice requires energy / latent heat to melt; [1 mark]
- This means it stays at 0°C / cooler for longer; [1 mark]

OR

This means it extracts more energy from the drink; [1 mark]

[Total: 2 marks]

4c

List the known quantities:

- Mass of plastic, $m_p = 0.210 \text{ kg}$
- Mass of silica gel and water solution, $m_s = 0.450 \text{ kg}$
- Specific heat capacity of plastic, $c_p = 1670 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of gel and water solution, $c_s = 6420 \text{ J kg}^{-1} \text{ K}^{-1}$
- Temperature change of plastic, $\Delta\theta_p = (19.0 - T_f)^\circ\text{C}$
- Temperature change of gel and water solution, $\Delta\theta_s = (T_f - 2.00)^\circ\text{C}$

Equate the amount of heat transferred by the plastic and silica gel water solution:

- The solution reaches thermal equilibrium when:
Heat supplied by the beaker = heat gained by the solution (since no heat is gained from or lost to the surroundings)

- $\Delta Q_p = \Delta Q_s$
- $m_p c_p \Delta\theta_p = m_s c_s \Delta\theta_s$
- $0.210 \times 1670 \times (19.0 - T_f) = 0.450 \times 6420 \times (T_f - 2.00)$ [1 mark]

Rearrange for the final temperature T_f :

- $350.7 \times (19.0 - T_f) = 2889 \times (T_f - 2.00)$
- $6663.3 - 350.7 T_f = 2889 T_f - 5778$
- $12441.3 = 3239.7 T_f$
- $T_f = 3.8403 = 3.84^\circ\text{C}$ (3 s.f.) [1 mark]

[Total: 2 marks]

Although the specific heat capacities have been given in units of $[\text{J kg}^{-1} \text{K}^{-1}]$, you do not need to convert the temperatures from $^\circ\text{C}$ to K. This is because the specific heat capacity equation requires the difference in temperature (ΔT) which is the same for either unit.

The temperature differences of the solution and the ice is calculated differently because:

- $\Delta\theta = T_f - T_i$ when $T_f > T_i$
- $\Delta\theta = T_i - T_f$ when $T_f < T_i$

4d

List the known quantities:

- Initial temperature, $T_f = 11.3^\circ\text{C}$ (from part c)
- Mass of plastic beaker, $m_p = 0.210 \text{ kg}$
- Mass of silica gel and water solution, $m_s = 0.450 \text{ kg}$
- Specific heat capacity of plastic beaker, $c_p = 1670 \text{ J kg}^{-1} \text{K}^{-1}$
- Specific heat capacity of gel and water solution, $c_s = 6420 \text{ J kg}^{-1} \text{K}^{-1}$
- Specific latent heat of fusion of ice, $l_i = 3.34 \times 10^5 \text{ J kg}^{-1}$
- Change in temperature of ice, $\Delta\theta_i = (0.9 - 0) = 0.9^\circ\text{C}$

- Change in temperature of the solution and plastic, $\Delta\theta_s = \Delta\theta_p = (11.3 - 0.9) = 10.4$ °C

Calculate the heat gained by the ice:

- Heat gained by ice = heat lost by plastic (due to the ice) + heat lost by solution (whilst the ice is melting)
- $\Delta Q_i = m_i c_w \Delta\theta_i + m_i l_i$
- $\Delta Q_i = [m_i \times 4200 \times 0.9] + [m_i \times (3.34 \times 10^5)]$ [1 mark]
- $\Delta Q_i = 3780 m_i + (3.34 \times 10^5) m_i = 337780 m_i$

Calculate the heat lost by the plastic beaker and gel-water solution:

- $\Delta Q_p + \Delta Q_s = m_p c_p \Delta\theta_p + m_s c_s \Delta\theta_s$
- $[0.210 \times 1670 \times 10.4] + [0.450 \times 6420 \times 10.4] = 33692.88 \text{ J}$ [1 mark]

Calculate the mass of ice added, m_i :

- $337780 m_i = 33692.88$
- $m_i = 0.0997 = 0.010 \text{ kg}$ [1 mark]

[Total: 3 marks]

This question can also be done by considering the solution returning back to its original (cooler) temperature. Therefore, the heat supplied by the plastic = heat used/gained by the ice.

5a

A specific heat capacity of $4200 \text{ J kg}^{-1} \text{ K}^{-1}$ represents:

- The energy required to raise 1 kg of water by 1 K (or °C); [1 mark]
- Is 4200 J; [1 mark]

[Total: 2 marks]

This question is asking about the specific heat capacity definition for water in particular, not just its general definition. At A level, relevant and clear explanations are required for full marks.

Calculate the heat gained by the ice:

- Heat gained by ice = heat lost by plastic (due to the ice) + heat lost by solution (whilst the ice is melting)
- $\Delta Q_i = m_i c_w \Delta \theta_i + m_i l_i$
- $\Delta Q_i = [m_i \times 4200 \times 0.9] + [m_i \times (3.34 \times 10^5)]$ [1 mark]
- $\Delta Q_i = 3780 m_i + (3.34 \times 10^5) m_i = 337780 m_i$

Calculate the heat lost by the plastic beaker and gel-water solution:

- $\Delta Q_p + \Delta Q_s = m_p c_p \Delta \theta_p + m_s c_s \Delta \theta_s$
- $[0.210 \times 1670 \times 10.4] + [0.450 \times 6420 \times 10.4] = 33692.88 \text{ J}$ [1 mark]

Calculate the mass of ice added, m_i :

- $337780 m_i = 33692.88$
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[Total: 3 marks]

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5a

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- The energy required to raise 1 kg of water by 1 K (or °C); [1 mark]
- Is 4200 J; [1 mark]

[Total: 2 marks]

This question is asking about the specific heat capacity definition for water in particular, not just its general definition. At A level, relevant and clear explanations are required for full marks.

5b

List the known quantities:

- Mass of water, $m = 0.75 \text{ kg}$
- Specific heat capacity of water, $c_w = 4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of ice, $c_i = 2.1 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific latent heat of fusion of ice, $l_i = 3.3 \times 10^5 \text{ J kg}^{-1}$

Total energy removed:

- Energy needed to cool water from 15°C to 0°C + energy needed for water to turn to ice + energy needed to cool ice from 0°C to -6°C

Energy needed to cool water from 15°C to 0°C :

- $\Delta Q_{\text{water}} = mc_w \Delta\theta$
- $\Delta Q_{\text{water}} = 0.75 \times (4.2 \times 10^3) \times (15 - 0) = 47\,250 \text{ J}$ [1 mark]

Energy needed from changing state from liquid (water) to solid (ice):

- $\Delta Q_{\text{change}} = ml_i$
- $\Delta Q_{\text{change}} = 0.75 \times (3.3 \times 10^5) = 247\,500 \text{ J}$ [1 mark]

Energy needed to cool ice from 0°C to -6°C :

- $\Delta Q_{\text{ice}} = mc_i \Delta\theta$
- $\Delta Q_{\text{ice}} = 0.75 \times (2.1 \times 10^3) \times (0 - (-6)) = 9\,450 \text{ J}$ [1 mark]

Total energy:

- $\Delta Q_{\text{water}} + \Delta Q_{\text{change}} + \Delta Q_{\text{ice}} = 47\,250 + 247\,500 + 9\,450 = 304\,200 \text{ J} = 0.3 \text{ MJ}$ [1 mark]

[Total: 4 marks]

In specific heat capacity questions, it helps to split the scenario into steps. In this question, the two steps are:

Water cools to 0°C → Water freezes to form ice → Ice cools to -6°C

Remember there is no temperature change when water is turning to ice (or anything is

changing state). However, energy is still required to change state.

5c

List the known quantities:

- Energy to remove 0.75 kg of water = 304 200 J (from part b)
- Time = 240 s

Calculate the energy to remove 1.25 kg of water:

- Energy to remove 1.25 kg of water = $\frac{1.25}{0.75} \times \frac{5}{3} \times 304\,200 = 507\,000 \text{ J}$ [1 mark]

Calculate power:

- The rate at which energy must be removed is the power
- Power = $\frac{\text{energy}}{\text{time}} = \frac{507\,000}{240} = 2112.5 = 2100 \text{ W}$ [1 mark]

[Total: 2 marks]

5d

Specific latent heat of vaporisation is greater than specific latent heat of fusion because:

Any **two** from:

- More energy has to be supplied to separate molecules (liquid to gas) than to break the solid bond between them (solid to liquid); [1 mark]
- For vaporisation, work is done against the atmospheric pressure; [1 mark]
- There is a greater change in potential energy from liquid to gas than from solid to a liquid; [1 mark]

[Total: 2 marks]

Latent heat of fusion is the amount of energy to change the state of 1 kg of liquid to solid **and** also from solid to a liquid. The same goes for latent heat of vaporisation (from liquid to gas or gas to liquid).